

VCAT

Home Project

Overview

The goal of this assignment is to create a small first-person turret shooting experience using Unity 2022.3.

The current project includes:

- Starter scene
- Several target objects
- Turret model
- Camera setup
- Basic placeholder scripts

You may freely refactor or replace any provided code if you believe it improves the solution.

Scope

The assignment is intentionally scoped to evaluate:.

- System architecture
- Code quality
- Extensibility
- Unity best practices

You are encouraged to prioritize:

- Data-driven design
- Simplicity and clarity
- Thoughtful architecture decisions

Gameplay Requirements

Input & Aim

- Turret aiming using mouse and/or keyboard input

Shooting

- Raycast-based shooting system
- Hit detection against targets
- Target destruction/deactivation on successful hit

Session Flow

- Session duration limit
- Limited ammunition / shot count
- Configurable active/inactive targets

Session End

- All active targets are destroyed
- Session timer expires
- Player runs out of shots

Configuration System

Designers should be able to create and switch between gameplay configurations without modifying code.

Each configuration should support:

- Session duration
- Maximum shot count
- Active/inactive targets

Preferred solutions are:

- Data-driven
- Easy to extend
- Inspector-friendly

Ballistics System

Our R&D team is currently developing a more realistic ballistics system intended to replace the existing raycast-based shooting implementation.

Please restructure the code in a way that allows the current implementation to be easily replaced with alternatives.

We expect the architecture to support:

- Minimal coupling between systems
- Straightforward replacement of the underlying firing implementation

Optional Features

If time permits, feel free to add:

- Sound effects
- Particle effects
- Main menu
- Sequential session configurations
- End-of-session statistics:
 - Total session duration
 - Accuracy
 - Targets hit
 - Shots fired

These additions are optional and will not outweigh code quality and architecture.

Evaluation Criteria

Code Quality

- Readability
- Maintainability

Architecture

- Modularity
- Appropriate use of Unity systems
- Data-driven approach

Technical Execution

- Correctness
- Stability
- Simplicity
- Performance awareness

Contact

For any questions or technical issues, feel free to contact:

ori.moskovitch@eyeviation.com

iliya.roginsky@eyeviation.com